Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-6 | Summer-2025

Course Code	:	2301CS601	**Date**	:	19-04-2025
Course Name	:	Operating System	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define the role of Operating System as a resource manager.	**4**
	(B)	Explain Monolithic Kernel with the help of diagram.	**3**
		OR	
		Explain Layered system with the help of diagram.	
	(C)	Define Operating System. Explain any five services/functions of Operating System.	**7**
		OR	
		Define Operating System. Explain any five types of Operating System.	
Q.2	**(A)**	Implement using Round Robin Scheduling algorithm. Consider Five Processes P1 to P5 arrived at same time. They have estimated running time 10, 2, 6, 8 and 4 seconds, respectively. Find the average turnaround time and average waiting time for Round Robin (quantum time=3) Scheduling algorithm. Ignore process swapping overhead.	**4**
	(B)	Explain Process States Transitions with the help of diagram.	**3**
		OR	
		Define Process Control Block (PCB). Explain different fields of Process Control Block (PCB).	
	(C)	Implement using SJF Scheduling algorithm. Five batch jobs A to E arrive at same time. They have estimated running times 10, 6, 2, 4 and 8 minutes. Determine mean process turnaround time for SJF algorithm.	**7**
		OR	
		Implement using non-pre-emptive priority Scheduling algorithm. Five batch jobs A to E arrive at same time. They have estimated running times 10, 6, 2, 4 and 8 minutes. Their priorities are 3, 5, 2, 1 and 4 respectively with 5 being highest priority. Determine mean process turnaround time for non-pre-emptive priority algorithm.	
Q.3	**(A)**	Define Deadlock. Explain necessary conditions for Deadlock.	**4**
	(B)	Explain Producer Consumer problem using Semaphore with algorithm.	**3**

OR

Explain Banker's algorithm for deadlock avoidance.

(C) Explain Readers & Writer Problem with algorithm. **7**

OR

Explain Dinning Philosopher Problem with algorithm.

Q.4 **(A)** Given six Partition of 300KB, 600KB, 350KB, 200KB, 750KB and 125KB(in order), solve how would the first-fit, best-fit and worst-fit algorithms places processes of size 115 KB, 500KB, 358KB, 200KB and 375KB. **4**

(B) Explain paging technique with the help of diagram. **3**

OR

What is Memory? Explain different types of memory.

(C) Consider the following reference string. Solve and calculate the page fault rates for FIFO page replacement algorithm. Assume the memory size is 4 page frame. **7**

1,2,3,4,5,3,4,1,6,7,8,7,8,9,7,8,9,5,4,5,4,2

OR

Consider the following reference string. Solve and calculate the page fault rates for LRU page replacement algorithm. Assume the memory size is 4 page frame.

1,2,3,4,5,3,4,1,6,7,8,7,8,9,7,8,9,5,4,5,4,2

Q.5 **(A)** Define RAID. Explain RAID0, RAID1 and RAID2. **4**

(B) Explain different file operations. **3**

OR

Explain different attributes of file.

(C) Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9, and 12, in that order. **7**
Starting from the current head position, solve and find the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for FIFO disk scheduling Algorithms?

OR

Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9, and 12, in that order.
Starting from the current head position, solve and find the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for SSTF disk scheduling Algorithms?

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